

SURAJ SINGH

Data Engineer | Big Data & Analytics

Python · SQL · PySpark · Snowflake · Microsoft Fabric · Databricks · Azure · AWS

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Professional Summary

Data Engineer with 1.5 years of experience designing cloud-native data pipelines, ETL/ELT workflows, and analytics solutions across Microsoft Fabric, Azure, and GCP. Proficient in Python, SQL, PySpark, Snowflake, and Databricks, with hands-on experience in data modeling, data warehousing, and Medallion architecture. Microsoft-certified (DP-700, DP-600) with a strong foundation in machine learning, NLP, and RAG-based AI systems. Adept at delivering production-grade data and AI solutions in collaborative, Agile enterprise environments.

Technical Skills

Languages: Python, SQL, C++, Java

Data Engineering & ETL: PySpark, Apache Spark, ETL/ELT, Data Pipelines, Data Modeling, Data Warehousing, Apache Kafka, Apache Airflow, dbt, SSIS, SSRS

Cloud & Data Warehouses: Microsoft Fabric, Azure, AWS, GCP, Snowflake, Databricks

Databases: PostgreSQL, MySQL, SQL Server (SSMS), MongoDB, Redis, Pinecone, Chroma

Machine Learning & AI: Machine Learning, NLP, LLMs, RAG Systems, Vector Search, Hugging Face, Sentence Transformers, Ollama

Analytics & Visualization: Power BI, Tableau, Looker, Streamlit, Data Analysis

Backend & DevOps: FastAPI, REST APIs, Microservices, Git, Docker, Podman, CI/CD, Linux

Professional Experience

Junior Data Engineer

Endava LLC

Feb 2025 – Present

Bangalore, India

- Engineered an AI-powered student application and course-recommendation platform on Microsoft Azure (Copilot Studio, Dynamics 365 CRM, Power Apps), building data-integration workflows across 5+ admission and enrollment processes that handled 10K+ student records and cut manual application-review effort by 35%.
- Migrated 15+ services of a grant-management platform to GCP, re-architecting Python data-processing and integration pipelines that reduced document-processing time by 45% and infrastructure costs by 20%, enabling AI-powered document analysis and workflow automation.
- Co-designed a modular AI guardrail framework adopted across 3+ enterprise initiatives, implementing deterministic validation pipelines with locally deployed LLMs (Ollama) that flagged 95%+ of sensitive PII/PHI and reduced data exposure across downstream flows.
- Developed backend integrations and cloud-native ETL workflows (Python, FastAPI, REST APIs) that ingested, transformed, and served 50K+ business records daily with 99%+ pipeline reliability to power analytics and reporting.
- Partnered with solution architects, platform engineers, and client stakeholders across Azure and GCP in Agile delivery cycles to ship production-ready data and AI solutions.

Machine Learning Engineer Intern

Ignitus

Sep 2023 – Mar 2024

Remote

- Built a custom text-generation system using Markov Models and Hidden Markov Models (HMMs) on 100K+ Reddit posts in Python, cutting manual content-generation effort by 30%.
- Developed an NLP sentiment-analysis pipeline processing 1,000+ URLs per run, engineering 10+ sentiment features with overall polarity scoring for downstream analytics.
- Designed reproducible machine learning pipelines processing 50K+ text records end-to-end, covering data ingestion, feature engineering, model evaluation, and experiment tracking.

Certifications

- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
- SnowPro Core Certified (Snowflake)
- Databricks Certified Data Engineer Associate
- AWS Certified Cloud Practitioner
- IBM Data Analytics; Infosys Machine Learning (Foundation & Advanced)

Key Projects

Manufacturing Analytics Platform <i>Python, SQL, Microsoft Fabric, AWS, Snowflake</i>	2026
- Architected a cloud-native analytics platform on Microsoft Fabric and Snowflake using Medallion Architecture (Bronze, Silver, Gold) to unify raw manufacturing data into analytics-ready layers. - Developed PySpark ETL pipelines and incremental-processing workflows, modeling Gold-layer tables consumed by Power BI dashboards for operational reporting.	
Alex RAG Assistant <i>Python, NLP, LLMs, Vector Search</i>	2026
- Built an LLM-powered document-intelligence platform (Python) for contextual question answering and knowledge retrieval over unstructured documents. - Implemented a RAG pipeline with semantic search using Sentence Transformers and a vector database (Pinecone/Chroma), delivering source-grounded responses to reduce hallucinations.	
Real-Time Market Data Pipeline <i>Kafka, Airflow, Snowflake, dbt</i>	2026
- Built a real-time market-data ingestion platform streaming from public APIs through Apache Kafka into Snowflake. - Orchestrated end-to-end workflows with Apache Airflow and implemented modular SQL transformations and testing in dbt for analytics-ready datasets.	

Education

Walchand Institute of Technology	2021 – 2025
B.Tech in Computer Science Engineering (Hons. in AI/ML)	CGPA: 9.5/10

Achievements

Smart India Hackathon Finalist – AI-Based Power System Vulnerability Detection
Publication: Scopus-Indexed Research Paper on ML-Based Legal Document Translation and Simplification